
UNIT 7 EVALUATION OF MULTIPLE REGRESSION MODELS*

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7.0 OBJECTIVES

After going through this Unit, you will be able to

- explain the concept of ‘Coefficient of Determination’ with an example;
- outline the procedure for testing the statistical significance of a single parameter in an estimated regression model;
- describe how the statistical significance of a group of parameters can be assessed by using the *F*-test; and
- Discuss the tests for linear restrictions, viz., Wald, Lagrange Multiplier, and Likelihood Ratio tests.

7.1 INTRODUCTION

In the previous Unit, we discussed the specification of multiple regression models. We also derived the estimators of the parameters by applying the OLS and maximum likelihood methods of estimation. In this unit, we will first consider the importance of the concept of Coefficient of Determination given by R^2 . We then proceed to learn how to test for the significance of the estimated parameters for individual regressors. We will also learn how to interpret the results. The extent to which the independent variables together explain the variation in the dependent variable will also be seen. We finally learn about three procedures for testing the linear restrictions.

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7.2 COEFFICIENT OF DETERMINATION

In Unit 4, while presenting the simple regression model, we derived the coefficient of determination, denoted by R -square (R^2). Recall that R^2 is defined as the ratio of ‘Explained Sum of Squares’ (ESS) to ‘Total Sum of Squares’ (TSS). In multiple regression models, we can define the coefficient of determination in the same manner.

Regression, as you know, is an analytical tool to explain the variation in the dependent variable (Y) in terms of the changes in the independent variables (X_i ’s). We can decompose the difference between Y_i and its arithmetic mean $\bar{Y}$ as follows:

$$(Y_i - \bar{Y}) = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \bar{Y}) \quad \dots (7.1)$$

Let us look into the intuitive idea behind the R^2 . The expected value of variable Y is $\bar{Y}$. Therefore, in the absence of the regression model, the expected value of Y is $\bar{Y}$. When we try to explain Y through a regression model, the expected value of Y is $\hat{Y}$. Therefore, the difference between $\hat{Y}_i$ and $\bar{Y}$, $(\hat{Y}_i - \bar{Y})$, is due to the explanatory variables included in the regression model and can be called as the ‘explained variation’. The difference $(Y_i - \hat{Y}_i)$ denotes the part that cannot be explained by the regression model. Hence, squaring both sides of equation (7.1) and summing over all observations (1 to n), we obtain

$$\sum (Y_i - \bar{Y})^2 = \sum (Y_i - \hat{Y}_i)^2 + \sum (\hat{Y}_i - \bar{Y})^2 \quad \dots (7.2)$$

Total Variation	Residual Variation	Explained Variation
Or, TSS = (Total Sum of Squares)	RSS + (Residual Sum of Squares)	ESS (Explained Sum of Squares)

R^2 is defined as

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} \quad \dots (7.3)$$

Note that R^2 measures the proportion of the variation in Y which is explained by the multiple regression equation. Therefore, R^2 is often used as a goodness-of-fit statistic to compare the validity of regression results under alternative specifications of the independent variables in the model.

MULTIPLE REGRESSION MODELS: TESTING

The value of R -square (R^2) ranges between 0 and 1. An implication of this bound is that R^2 is always positive. Let us consider two hypothetical situations: First, when $ESS = TSS$ (i.e., $RSS = 0$) when the explanatory variables explain the variation in the dependent variable fully. Then, $R^2 = 1$. Second, when $ESS = 0$ (i.e., the explanatory variables do not explain any variation in Y). Then, $TSS =$

RSS and $R^2 = 0$. Thus, the range within which the coefficient of determination remains is $0 \leq R^2 \leq 1$.

The coefficient of determination (R^2) reflects the explanatory power of a regression model. A higher value of R^2 implies higher explanatory power of the underlying regression model compared to some other model. Thus, R^2 can be used as a measure of goodness of fit in multiple regression models. Remember that we can compare two values of R^2 only when the dependent variable is the same. *We cannot compare two regression models if the dependent variable is different.*

On re-arrangement of terms in equation 7.3, we obtain

$$R^2 = \frac{ESS}{TSS} = \frac{TSS - RSS}{TSS} = (1 - \frac{RSS}{TSS})$$

Thus,

$$R^2 = 1 - \frac{\sum \hat{u}_i^2}{\sum (Y_i - \bar{Y})^2} \quad \dots (7.4)$$

From (7.4) we find that as RSS decreases, R^2 increases.

7.2.1 Over-Fitting

Suppose we have a regression model with two explanatory variables, and we obtain R^2 value as 0.63. It implies that the two explanatory variables explain 63 per cent variation in the dependent variable. Now, suppose, we add one more explanatory variable to the regression model, i.e., the number of explanatory variables is three. What will happen to the R^2 value? We can compare the R^2 values of both the regression models since the dependent variable is the same.

It is observed that when we add explanatory variables to a regression equation, the R^2 value increases (i.e., the explanatory power of the model increases). The R^2 value never decreases when we add an explanatory variable. Thus, there is a tendency on the part of researchers to include more explanatory variables in a regression model. In such cases, even if logically not correct and statistically not significant, many explanatory variables get added to the regression model. We cannot trust the inferences drawn from such models! Therefore, a question arises is: where do we stop?

7.2.2 Adjusted- R^2

The degrees of freedom of a regression model with sample size n and number of explanatory variables k (including the intercept) for the three sums of squares are as follows:

$$\text{TSS: } (n - 1) \qquad \text{ESS: } 1 \qquad \text{RSS: } (n - k)$$

There is thus a decrease in the degrees of freedom of RSS when we add explanatory variables to a model. Notice that R^2 as a measure of goodness of fit does not account for the degrees of freedom. A natural solution is to use *variances* (i.e., sums of squares divided by degrees of freedom), and not mere *variations*. It eliminates the dependence of goodness of fit on the number of

independent variables in the model. Such an R^2 is defined as $\bar{R}^2$, and is called as the corrected/adjusted R^2 . In notations, we have

$$\bar{R}^2 = 1 - \frac{RSS/(n-k)}{TSS/(n-1)} \quad \dots (7.5)$$

On re-arrangement of terms in equation (7.5), we obtain

$$\bar{R}^2 = 1 - \frac{RSS (n-1)}{TSS (n-k)} = 1 - \frac{(TSS - ESS) (n-1)}{TSS (n-k)} = 1 - \left[\left(1 - \frac{ESS}{TSS} \right) \frac{(n-1)}{(n-k)} \right]$$

Or,

$$\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)} \quad \dots (7.6)$$

where n is number of observations in the regression model and k is number of parameters in the equation. As n becomes large, the difference between $\bar{R}^2$ and R^2 tend to be small.

$\bar{R}^2$ has several properties. One, when new variables are added to a regression model, the R^2 always increases, but the $\bar{R}^2$ may rise or fall. Second, the use of $\bar{R}^2$ eliminates the incentives for researchers to include more variables in a model without logic. Third, $\bar{R}^2$ is always less than or equal to R^2 . Fourth, the $\bar{R}^2$ can also be negative while R^2 is always positive. For these reasons, $\bar{R}^2$ is regarded as a more desirable measure of goodness of fit compared to R^2 . Let us now consider an example on multiple regressions.

Example 7.1:

Consider the regression model $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + u$. The following data are given. Calculate the values of R^2 and $\bar{R}^2$.

Y	40	60	50	70	90
X_1	4	6	7	10	13
X_2	8	12	10	5	7

We prepare a table as below to find out the arithmetic mean of Y , X_1 and X_2 . Then, we subtract the mean from the respective variables to obtain y , x_1 and x_2 .

Y	X_1	X_2	y	x_1	x_2
40	4	8	- 22	- 4	- 0.4
60	6	12	- 2	- 2	3.6
50	7	10	- 12	- 1	1.6
70	10	5	8	2	- 3.4
90	13	7	28	5	- 1.4

From the above table we obtain the following:

$$\begin{array}{lll} \sum Y = 310 & \sum X_1 = 40 & \sum X_2 = 42 \\ \sum x_1 y = 260 & \sum x_2 y = -84 & \sum x_1 x_2 = -21 \\ \sum x_1^2 = 50 & \sum x_2^2 = 29.70 & \sum y^2 = 1480 \\ n = 5 & & \end{array}$$

Therefore,

$$\begin{aligned} \hat{\beta}_1 &= 5.72 \\ \hat{\beta}_2 &= 1.24 \\ \hat{\alpha} &= 5.86 \end{aligned}$$

Further,

$$RSS = \sum \hat{u}^2 = 96.84$$

The standard error of the estimators are

$$\begin{aligned} s_{\hat{\alpha}} &= 6.96 \\ s_{\hat{\beta}_1} &= 1.18 \\ s_{\hat{\beta}_2} &= 1.54 \end{aligned}$$

Thus, the estimated equation is

$$\hat{Y} = 5.86 + 5.72 X_1 + 1.24 X_2$$

From this equation we can compute

$$\begin{aligned} ESS &= \sum \hat{y}^2 = 1383.16 \\ TSS &= \sum y^2 = 1480.00 \\ R^2 &= \frac{ESS}{TSS} = 0.93 \end{aligned}$$

Adjusted- R^2 is

$$\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)}$$

Or,

$$\bar{R}^2 = 1 - (1 - 0.93) \frac{(5-1)}{(5-3)} = 1 - 0.07 \times \frac{4}{2} = 0.86.$$

Check Your Progress 1

- 1) Define the Coefficient of Variation. Indicate its expression for multiple regression models.

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- 2) Why is R^2 often used as a measure of ‘goodness of fit’?
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- 3) Specify an alternative expression for R^2 indicating its relationship with the Residual Sum of Squares. What is the relationship that is conveyed by this expression?
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- 4) Is it always beneficial to add another ‘explanatory variable’ in a multiple regression model? What is a way out of this situation?
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- 5) Explain why the adjusted- R^2 is preferred to R^2 .
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7.3 HYPOTHESIS TESTING

In the previous Unit, we learnt that under the Gauss-Markov assumptions, the OLS estimator b has normal distribution with a mean 0 and covariance matrix $\sigma^2(X'X)^{-1}$. We can use this information to test the hypotheses associated with the unknown population parameter vector β . For testing the significance of b_k we will use the test statistic:

$$z = \frac{b_k - \beta_k}{\sigma \sqrt{c_{kk}}} \quad \dots (7.7)$$

where z is the test statistic;

β_k is the parameter to be tested;

b_k is the estimate of β_k obtained from the sample data; and

c_{kk} is the standard error of b_k .

The variance-covariance matrix of the estimators is given by $\text{var}(\mathbf{b}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. To obtain the standard error of the estimators, we first compute the inverse of the matrix $(\mathbf{X}'\mathbf{X})$. Second, we multiply it by σ^2 . Third, we take the square root of the matrix $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. Suppose, the ij^{th} element of this matrix is denoted by σc_{ij} . The diagonal elements provide us the standard error of the individual parameters. For example, the first diagonal element (let us say, σc_{11}) provides us with the standard error of b_1 . Similarly, the second diagonal element (σc_{22}) provides us with $se(b_2)$. Thus, when we test for the statistical significance of b_k , we should take its standard error $\sigma \sqrt{c_{kk}}$ as denominator.

The test statistic z follows standard normal distribution. If we replace σ with its estimate s , it can be shown that the unbiased estimator s^2 is independent of b and follows chi-squared distribution with $(n - k)$ degrees of freedom. The corresponding test statistic is given as

$$\chi^2_{(n-k)} \sim \frac{(n-k)s^2}{\sigma^2} \quad \dots (7.8)$$

We can, therefore, define the random variable t_k to test for b_k as

$$t_k = \frac{b_k - \beta_k}{s \sqrt{c_{kk}}} \quad \dots (7.9)$$

The value of the t -ratio in equation (7.9) is the ratio of a standard normal variable $(b_k - \beta_k)$, and the square root of an independent chi-squared variable $(s \sqrt{c_{kk}})$. It therefore follows the Student's t -distribution with $(n - k)$ degrees of freedom. The t -distribution (see Appendix - B at the end of the book) resembles the standard normal distribution for large values of $(n - k)$. For very large values of $(n - k)$, the two (t -distribution and z -distribution) are identical. The expression (7.9) can be used to further develop 'test statistic' and 'confidence interval'.

7.3.1 t -Test

Testing of a hypothesis involves two steps: (i) construction of a test statistic, and (ii) a decision rule. Let me illustrate it this with an example. Suppose I obtained the estimate b_k . I want to test the hypothesis $H_0: \beta_k = 0$. The alternative hypothesis is $H_1: \beta_k \neq 0$. It implies that I have a two-tailed test. From (7.9) I have the test statistic

$$t_k = \frac{b_k - 0}{se(b_k)} = \frac{b_k}{se(b_k)} \quad \dots (7.10)$$

In (7.10), there are no unknown values: b_k is the estimate obtained from the data; and $se(b_k)$ is the standard error of b_k that is computed from the dataset. Hence, I can compute the test statistic t_k by taking the ratio of b_k and $se(b_k)$.

The strategy to test the null hypothesis is as follows: I consider the degrees of freedom $(n - k)$, and the level of significance α . I compare the calculated t -value (obtained from equation (7.10) with the tabulated t -value (given at Appendix B). I compare the absolute value of t_k with the tabulated value. If the calculated value of t_k exceeds the tabulated value, I reject the null hypothesis. If the calculated value of t_k is less than the tabulated value, I *do not* reject the null hypothesis.

The critical value of t is defined as

$$P\{|t_k| > t_{n-k;\alpha/2}\} = \alpha$$

The critical value for $\alpha = 0.05$ for very large samples is 1.96. Thus, at the 5% level of significance the null will be rejected if

$$|t_k| > 1.96$$

Since under the alternative hypothesis, we have values for β_k on both sides of β_k^0 ($H_0: \beta_k = \beta_k^0$) the test is referred to as a two-tailed test. In case we have the null hypothesis as $H_0: \beta_k \leq \beta_k^0$ with an alternative hypothesis $H_1: \beta_k > \beta_k^0$, we reject the null hypothesis if t_k is too large. However, large negative values for t_k does not lead to rejection of the null hypothesis. This is known as one-sided test and the critical value is determined by

$$P\{t_k > t_{(n-k);\alpha}\} = \alpha$$

The critical value for $\alpha = 0.05$ for such a one-tailed test is 1.64. Thus, at the 5% level of significance the null will be rejected if

$$t_k > 1.64$$

Most of the standard statistical software reports the t -ratio as point estimates divided by its standard error. The t -ratio refers to the t statistics that is computed to test the hypothesis that $\beta_k = 0$. Rejection of the null hypothesis implies that the estimate b_k differs significantly from 0; or that the corresponding regressor x_{ik} has significant impact on y_i .

7.3.2 Confidence Interval

A confidence interval is the interval consisting of all values for β_k^0 for which the t -test does not reject the null hypothesis that $\beta_k = \beta_k^0$. It provides a range of values for true β_k given the estimate b_k and its standard error. Thus, for the given probability $(1 - \alpha)$ it implies that

$$-t_{(n-k);\alpha/2} < \frac{b_k - \beta_k^0}{se(b_k)} < t_{(n-k);\alpha/2} \quad \dots (7.11)$$

This can be written as

$$[b_k - t_{(n-k);\alpha/2} se(b_k)] < \beta_k < [b_k + t_{(n-k);\alpha/2} se(b_k)] \quad \dots (7.12)$$

Finally, using standard normal approximation, a 95% confidence interval for β_k is given by

$$[b_k - 1.96 se(b_k)] < \beta_k < [b_k + 1.96 se(b_k)] \quad \dots (7.13)$$

This implies that, in repeated sampling, in 95% cases, these intervals will contain true value of β_k .

7.3.3 F-Test

Almost all the statistical packages provide the results for the joint test of significance of the regression coefficients. It tests the hypothesis that all the coefficients except the intercept are equal to zero. The null here is that J of the k coefficients ($J < k$) are equal to zero.

$$H_0: \beta_{k-J+1} = \dots = \beta_k = 0 \quad \dots (7.14)$$

The alternative hypothesis is that at least one of these coefficients is not equal to zero. The test procedure applied here involves comparing the Residual Sum of Squares of the full model denoted by RSS_1 with Residual Sum of Squares (RSS) of the restricted model (model with the last J regressor omitted) denoted by RSS_0 . A test statistic for this can be obtained by using the following rationale.

Under the null hypothesis, and the assumptions discussed in Section 6.5 of the previous Unit, we have

$$\chi_J^2 \sim \frac{RSS_0 - RSS_1}{\sigma^2} \quad \dots (7.15)$$

From the null hypothesis, we know that $(RSS_0 - RSS_1)$ and s^2 (which is an estimator of σ^2) are independent and hence the test statistic can be defined as

$$F = \frac{\frac{(RSS_0 - RSS_1)}{J}}{\frac{s^2}{(n-k)}} \quad \dots (7.16)$$

The test statistic F is distributed as F with J and $(n-k)$ degrees of freedom. This is denoted as $F_{(n-k)}^J$. Hence, the test is also known as F -test. Using the definition of R^2 we can write (7.16) as

$$F = \frac{\frac{(R_1^2 - R_0^2)}{J}}{\frac{(1 - R_1^2)}{(n-k)}} \quad \dots (7.17)$$

where R_1^2 and R_0^2 are the goodness-of-fit measures for the unrestricted model and the restricted model respectively. The rejection of the null hypothesis is possible only in the case of large values of the test statistic. Although the alternative hypothesis is two-sided, based on the distribution, the critical values $F_{n-k; \alpha}^J$ of F is one-sided and is defined as

$$P\{F > F_{(n-k); \alpha}^J\} = \alpha, \text{ for the level of significance } \alpha.$$

A special case of F -test is where we test the null hypothesis that all the partial slope coefficients are zero, i.e., $H_0: \beta_2 = \beta_3 = \dots = \beta_k = 0$. In this case, the test statistic is given as

$$F = \frac{\frac{(RSS_0 - RSS_1)}{(k-1)}}{\frac{RSS_1}{(n-k)}} \quad \dots (7.18)$$

where RSS_0 is the RSS of the restricted model and RSS_1 is the RSS of the unrestricted model. Further, in this case, since in the restricted model only the intercept term is present, the test statistic using R^2 values is given as

$$F = \frac{\frac{R^2}{(k-1)}}{\frac{(1-R^2)}{(n-k)}} \quad \dots (7.19)$$

Here, F is used to denote the test statistic expressed as a simple function of R^2 . This implies that if the test based on F fails to reject the null hypothesis, we can infer that the model performs poorly. However, if the null hypothesis is rejected, we cannot infer that the model performs better.

Check Your Progress 2

- 1) Which ‘statistic’ is used to test for the significance of estimated individual regression coefficients? How is the test result arrived at?
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- 2) How is the rejection of a null hypothesis interpreted?
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- 3) Define the ‘confidence interval’ of a parameter. What does it imply?
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- 4) When is the F -statistic used? Indicate its expression in terms of R^2 stating how the interpretation is made depending on its value obtained for a sample.
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7.4 TESTING LINEAR RESTRICTIONS

The t -test discussed in Sub-Section 7.3.1 involved restriction on one parameter only. In practise, however, we come across situations which requires linear restrictions on more than one parameter such as $\beta_2 + \beta_3 + \dots + \beta_k = 1$.